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2024 APMCM summary sheet

As pet demand soars, the industry faces challenges and opportunities in standardization and supply chain diversification, driven by the pursuit of high-quality development and new regulations. Our study employs ARIMA, Grey Prediction Model GM(1,1), and polynomial regression to analyze and forecast the pet industry's complex dynamics, providing strategic decision support.

Question 1 necessitates a forecast of the developmental trends of China's pet industry over the next three years. By scrutinizing data spanning the past five years of China's pet industry, we have employed ARIMA and the Grey Prediction Model GM(1,1) to project future trajectories. The analysis indicates a sustained increase in the population of pet cats, juxtaposed with a marginal decline in the population of pet dogs.

Question 2 calls for a projection of global demand for pet food over the ensuing three years. We have utilized grey relational analysis from Grey System Theory and the GM(1,1) model to dissect the development of the global pet industry and to anticipate global demand for pet food. The results reveal an overarching upward trend in the global pet population, with a concomitant escalation in demand for pet food.

Question 3 analyzes China's pet food industry trends and market demand, unaffected by economic policies. Using Grey Prediction and polynomial regression models, we forecast stable growth in production and exports for the next three years. The forecasts suggest that the production and export of pet food in China are poised to continue on a stable growth path.

Question 4 is tasked with examining the impact of novel foreign trade policies from European nations and the United States on China's pet food industry. We have constructed multiple linear regression models to quantify the effects of these new trade policies on China's pet food industry and have formulated strategies for the sustainable development of China's pet food industry based on the outcomes of our model analyses.

This paper innovates by integrating models to analyze the global pet industry, focusing on China and international markets, and offering decision support for the industry's sustainable development, with models adaptable to limited data.

Keywords: Global Pet Industry Pet Food Sector ARIMA Modeling Grey Prediction Model GM(1,1) Modeling Polynomial Regression Linear Regression

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I. Introduction

1.1 Question Background

The global pet industry has experienced sustained growth, albeit at a slower pace, and has seen its market size expand. Key drivers of this growth include the increasing demand for pets as companions, particularly among urban youth, driven by a demographic shift towards an aging and growing single population [1]. Furthermore, the pursuit of high-quality social development and the implementation of new regulations have contributed to the standardization of the industry.

As the industry continues to evolve, its supply chain becomes increasingly diversified, encompassing breeding, sales, food, and services, with the pet food sector emerging as a significant segment. This evolution has given rise to a trend towards specialized infrastructure and digital management, offering both challenges and opportunities for businesses. To address these complexities, mathematical modeling can be employed to predict market trends and inform strategic decision-making.

This paper aims to develop mathematical models that analyze trends within the pet industry and quantify complex phenomena, with a particular focus on mammals and pet food. These models will utilize historical and current data to forecast future trends, thereby supporting strategy formulation and decision optimization. They will streamline problem-solving, validate assumptions, evaluate risks, optimize resource allocation, and capture the dynamics of the industry, serving as a powerful tool for enhancing our understanding and advancing the development of the pet industry development.

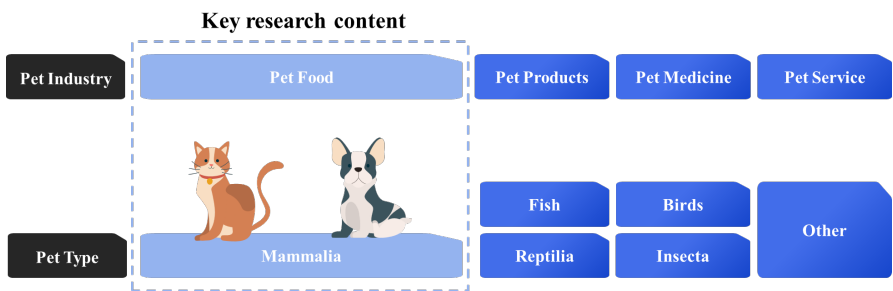


Figure 1 Pet Industry Key Research Content

1.2 Restatement of Problem

Question 1: Based on the data in Attachment 1 and the additional data collected by your team, please analyze the development of China’s pet industry in the past five

years by pet type. And analyze the factors in the development of China's pet industry, so as to make an appropriate mathematical model to predict the development of China's pet industry in the next three years.

Question 2: Overseas pet industries, such as in European countries and America, have also developed rapidly in recent years. Please analyze the development of the global pet industry by pet type, based on the data in Attachment 2 and the additional data collected by your team. And make an appropriate mathematical model to forecast the global demand for pet food in the next three years.

Question 3: In light of China's pet food production and export values in Attachment 3, please analyze the development of China's pet food industry and predict its production and export of pet food in the next three years (regardless of changes in economic policies), based on the trend of global demand of pet food market and China's development

Question 4: China's pet food industry will inevitably be influenced by the new foreign economic policies (such as tariff policies) of European countries and America. To quantitatively analyze this influence, please make an appropriate mathematical model and take the data in the attachments, the additional data you collected, and the calculation results in above questions into consideration. Based on your calculations, please develop feasible strategies for the sustainable development of China's pet food industry.

II. Problem analysis

2.1 Analysis of Problem One

To analyze the development of China's pet industry over the past five years and forecast trends for the upcoming three years, we initially compile the dataset and subsequently perform descriptive statistical analysis to discern the temporal patterns in pet population and market sales.

Utilizing these analyses, we develop a mathematical model based on time series analysis, specifically the Autoregressive Integrated Moving Average (ARIMA) model, which accounts for various influencing factors and their interrelationships [?]. Once the model is developed, we will employ historical data to validate and refine the model, thereby enhancing the precision of our predictions.

Ultimately, we will apply the refined model to project the trajectory of China's pet industry over the subsequent three years, encompassing the anticipated growth in pet populations and fluctuations in market sales.

2.2 Analysis of Problem Two

The growth of the global pet population is a complex system influenced by a multitude of factors, spanning social, economic, and cultural aspects, with a certain degree of uncertainty and nonlinear relationships among the data. To address these uncertainties and nonlinearities, we employ the Grey Correlation Analysis and the Grey Forecasting Model from Grey System Theory to handle such uncertainties and reveal the intrinsic connections between various factors [2]. Moreover, unlike traditional statistical models such as regression analysis, the Grey Forecasting Model does not require a large amount of historical data. It has lower demands for historical data and can predict future trends based on a limited set of observational data [3]. Therefore, it is particularly suitable for situations where extensive historical data is scarce.

2.3 Analysis of Problem Three

Leveraging the projected global demand for pet food as established in the preceding analysis and the production and export data of China's pet food industry detailed in Appendix 3, we have developed a Grey Prediction Model and a Polynomial Regression Model. These models are designed to anticipate the production and export volumes of China's pet food over the forthcoming three years. Through an examination of the trajectory of China's pet food industry, we project the industry's production and export dynamics for the subsequent three years, under the assumption of stability in economic policies.

2.4 Analysis of Problem Four

To conduct a comprehensive analysis of the effects of recent foreign economic policies on China's pet food industry, we have formulated multiple linear regression models. These models are based on data sourced from the attachment, supplementary data collection, and the outcomes of prior analyses. They evaluate the impact of varying tariff policies on the industry. Utilizing the insights gleaned from the model analysis, we have developed a suite of viable strategies aimed at fostering the sustainable development of China's pet food industry. This includes strategic adaptations to policy alterations and initiatives for market diversification.

III. Model Hypothesis

3.1 Assumption One

Data is error-free, independent, and stable: We presume data accuracy and independence across years, with short-term stability for trend prediction based on the last five years.

3.2 Assumption Two

Pet market is unsaturated, and production is scalable: We assume room for growth in the pet food market and scalability of production to meet increasing demand.

3.3 Assumption Three

Consumer spending patterns are stable: Assuming no major social, economic, or cultural shifts, consumer demand for pet food remains stable for short-term forecasting.

3.4 Assumption Four

Exchange rates remain stable during analysis: We assume stable exchange rates to avoid significant impacts on international market analysis and pet food price competitiveness.

IV. Notations

Table 1 Description of Time Series Data Parameters

Parameter	Description
YEAR	A vector containing the years
a	Growth rate parameter in the model
u	Compensatory term or constant in the model
x	year

V. Model Establishment and Solution of Question 1

5.1 Data Source and Preprocessing

We have curated data pertaining to China's pet industry from 2019 to 2023, expressed in units of ten thousand, encompassing the populations of cats and dogs. The data is sourced from Appendix I, ensuring its accuracy and reliability. Furthermore, we conducted a separate analysis of the growth rates for the cat and dog populations:

$$\text{Growth Rate} = \left(\frac{\text{Current Year's Quantity} - \text{Previous Year's Quantity}}{\text{Previous Year's Quantity}} \right) \times 100\%$$

Utilizing the growth rate formula, we calculated the growth rates for the pet populations over the past four years. Subsequently, this data was visualized, resulting in the stacked column chart presented in Figure 2.

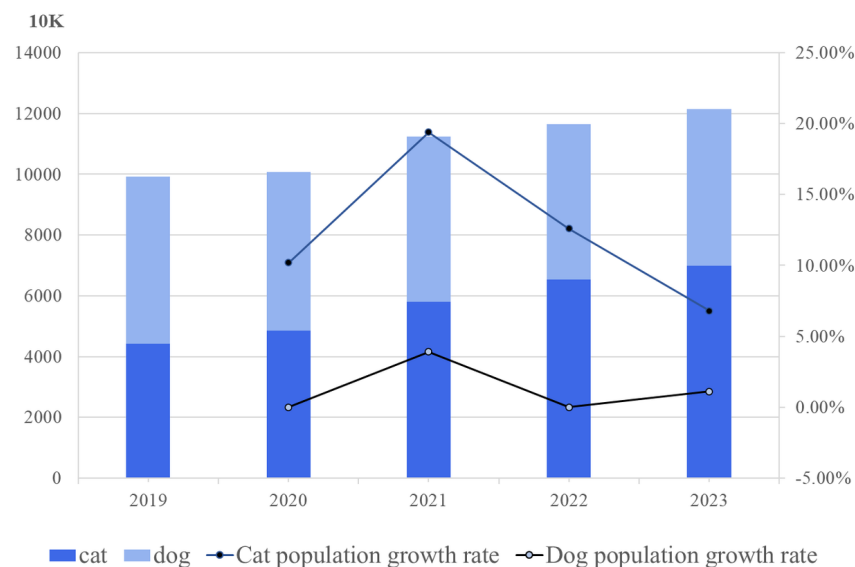


Figure 2 Number of pet cats and dogs in China, 2019-2023

Figure 2 reveals distinct characteristics in the population data of cats and dogs. The cat population data demonstrate a significant upward trend, whereas the dog population data exhibit greater stability. Given these observations, we have chosen to initially apply an ARIMA model to the cat population data and employ first-order differencing to render the series stationary. This preparation is essential for the development of ARIMA models. Concurrently, the Grey Forecasting Model is applied to the dog population data.

5.2 Model Selection and Parameter Optimization

5.2.1 *Optimal model selection: Cat*

Initially, we performed a stationarity test on the cat population data, which indicated non-stationarity. Consequently, we applied first-order differencing to the cat population data to induce stationarity. Thereafter, we specified an ARIMA model to model the cat population data, incorporating autoregressive (AR), integrated (I), and moving average (MA) components, with the differencing parameter set to 1. To ascertain the optimal ARIMA model, we explored all permutations of AR and MA orders, with both orders capped at 2, and estimated and assessed each model permutation to ensure comprehensive consideration of all potentially optimal solutions [4]. As a result, we obtained the ARIMA model coefficients as shown in Table 2.

The evaluation criteria for the model are as follows:

- **T-Statistic:** Measures the degree of difference between the coefficient and zero. A larger absolute value indicates that the coefficient is significantly different from zero.
- **P-Value:** Associated with the T-Statistic, it represents the probability that the coefficient is significantly different from zero. A smaller P-value (typically less than 0.05) indicates that the coefficient is significant.

Upon comparison, the ARIMA(0,1,0) model emerged as the most suitable, as its coefficients lacked significance, suggesting an absence of overfitting and offering a concise and efficacious model for data representation.

5.2.2 *Optimal model selection: Dog*

For dog population forecasting, we opted for the Grey Forecasting Model GM(1,1), given its significant predictive efficacy with a limited number of data points. To validate the appropriateness of the dog population data for grey model construction, we performed a ratio-to-interval test on the time series data. All ratio values from the original sequence were within the interval (0.717, 1.396), confirming the data's suitability for grey model development.

5.2.3 *Model Establishment*

A. ARIMA Model

Table 2 ARIMA Model Coefficients

Model	Parameter	Value	Standard Error	TStatistic	PValue
ARIMA(0,1,0)	Constant	-0.84193	365.82	-0.002299	0.99817
	Variance	1.8581e+05	3.1634e+05	0.58738	0.55695
ARIMA(0,1,1)	Constant	10	101.22	0.098796	0.9213
	MA[1]	-0.88559	1.2549	-0.70573	0.48836
	Variance	1.2299e+05	1.6276e+05	0.7501	0.4532
ARIMA(0,1,2)	Constant	10	441.95	0.022627	0.98195
	MA[1]	-0.47658	22.04	-0.021624	0.98275
	MA[2]	-0.52342	10.632	-0.04923	0.96074
	Variance	57064	1.4782e+05	0.38604	0.69947
ARIMA(1,1,0)	Constant	10	1.3814	7.2392	4.5147e-13
	AR[1]	0.99983	0.004524	221	0
	Variance	1.0001	0.010076	99.263	0
ARIMA(1,1,1)	Constant	-4.8616	39.724	-0.12239	0.90259
	AR[1]	-0.89183	0.94206	-0.94668	0.3438
	MA[1]	-1	1	-0.99997	0.31732
	Variance	30007	69378	0.43251	0.66537
ARIMA(2,1,0)	Constant	7.8822e-07	3.7765	2.0872e-07	1
	AR[1]	-0.97196	6.5183	-0.14911	0.88146
	AR[2]	-1	0.017562	-56.941	0
	Variance	0.0015775	0.0031811	0.4959	0.61996

P-values less than 0.05 indicate statistical significance.

The ARIMA(0,1,0) model, at its core, is a random walk model that attains stationarity via a single differencing process. To enhance simplicity and interpretability, we employed linear regression to approximate the ARIMA(0,1,0) model. Within this linear regression framework, the year is treated as the independent variable, while the cat population time series values constitute the dependent variable. The model parameters were estimated employing the least squares method, resulting in the following regression equation:

$$Y = -1006981.8 + 501.0 \times \text{YEAR}$$

Following the derivation of the equation, we conducted a linear regression analysis to evaluate the model's adequacy, yielding the following outcomes:

- **F-Statistic:** The F-test demonstrates the model's significance ($F=13.063$, $P=0.036$), permitting the rejection of the null hypothesis that there is no linear relationship.
- **R² (R-squared):** An R-squared value of 0.813 suggests a good fit of the model to the data.
- **Variance Inflation Factor (VIF):** A VIF value of 1 for the year variable indicates the absence of multicollinearity.

B. Grey Forecasting Model (GM(1,1))

Table 3 Fitting Results of GM(1,1) Model

Year	Actual Value	Predicted Value	Relative Error (%)
2020	5222	5303.669	1.564
2021	5429	5258.447	3.142
2022	5119	5213.611	1.848
2023	5175	5169.157	0.113

Note: The data unit is ten thousand.

The Grey Forecasting Model is a crucial element of Grey System Theory, enabling the construction of mathematical models and predictions from limited and incomplete data. Given our focus on Grey System Theory in Problem 2, further elaboration is reserved for that section. Utilizing historical data, we developed the GM(1,1) model for dog population forecasting and conducted a post-forecast difference test, yielding a post-forecast error rate of 0.4%. This suggests that, despite the dispersion in the original dog population data, the forecasted values align closely with the actual figures,

indicating an acceptable level of model accuracy. The grey forecasting model is defined as:

$$x^{(0)}(k) = \left(x^{(0)}(1) - \frac{u}{a}\right) e^{-a(k-1)} + \frac{u}{a} \quad (k = n+1, n+2, \dots, n+n_{\text{forecast}})$$

the parameters are as follows: a is -0.1149, u is 4250.8302.

Furthermore, we compared the predictions with the actual values from 2020 to 2023. The relative error ranged from 0.113% to 3.142%, with an average of 1.333%, demonstrating a good fit, as detailed in Table 3.

5.3 Model Prediction Results and Analysis

And the formula for Dog population growth is as follows:

By employing the ARIMA model to forecast the number of pet cats and the Grey Prediction Model GM(1,1) for the number of pet dogs, we can plot the fitting curve as shown in Figure 3. According to the chart, it is not difficult to discern the predicted trends for the number of pet cats and dogs:

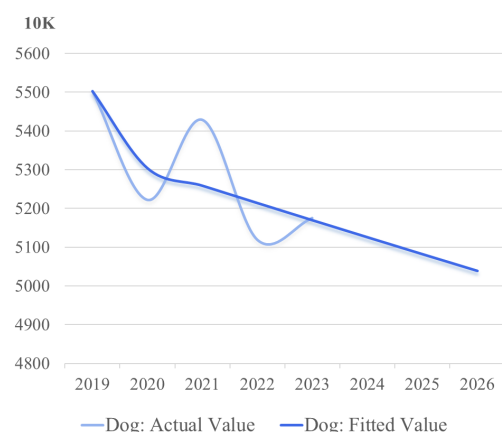
- **Pet Cat Quantity Forecast:** The ARIMA model predicts that from 2023 to 2026, the number of pet cats is expected to continue to grow, increasing from 70.42 million to 80.44 million. This growth trend aligns with the continuous increase in the number of cats from 2019 to 2023, indicating a significant rise in the popularity of cats as pets.
- **Pet Dog Quantity Forecast:** According to the predictions from the Grey Prediction Model GM(1,1), from 2024 to 2026, the number of pet dogs will show a declining trend, reducing from 51.25 million to 50.38 million, but the decline is slow and the overall trend is relatively stable. This trend corresponds with the relatively stable number of pet dogs from 2019 to 2023, suggesting that the pet dog market may be approaching saturation.

From 2022, cat ownership has exceeded dog ownership, reflecting a shift in the pet market and increasing preference for cats. Figure 2 shows a higher growth rate for cats. Several factors contribute to this trend:

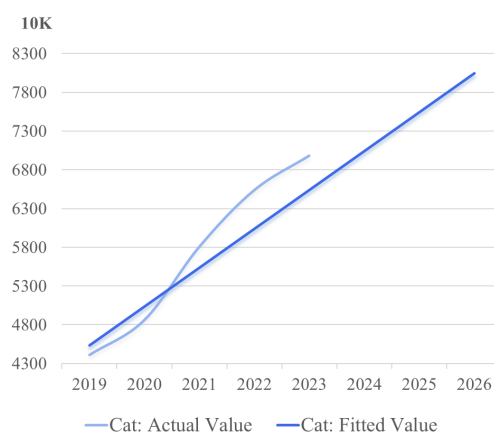
- **Demographic Changes:** The "China Pet Industry Development Index Report (2023)" links the pet industry's growth to societal factors like aging and single-person households, where pet ownership is seen as easier than raising children.
- **Economic Development:** Higher income levels, as noted in the "2021 China Pet Industry White Paper," have increased the consumption capacity and willingness

of pet owners, leading to market growth.

- **Technological Empowerment:** Technology has improved pet product and service quality, with the "2024 China Pet Industry Research Report" highlighting that cat owners can afford better care due to lower food costs compared to dog owners.
- **Policy Support:** National policies have fostered a favorable environment for the pet industry's development, as shown in the "China Pet Industry Development Index Report (2023)."



(a) The Development of Pet Dogs



(b) The Development of Pet Cats

Figure 3 Development of Pet Dogs and Cats over the Next Three Years

Forecasts suggest that cat numbers will keep rising while dog numbers will decline. The pet market will see:

- **Market Size Growth:** Continued expansion, with cats outnumbering dogs, aligning with the industry's transformation towards openness and prosperity.
- **Product and Service Innovation:** Diversification to meet personalized demands, reflecting consumer trends.
- **Personalization and Professionalization of Pet Services:** Services will become more segmented and professional, aiming to enhance pet quality of life.
- **Industry Standardization and Regulation:** The industry will become more standardized, reflecting professional development trends across the pet industry chain.

VI. Model Establishment and Solution of Question 2

6.1 Data Source and Preprocessing

We utilized the data from Appendix 2, which includes the number of different regions and pet species from 2019 to 2023 (unit: ten thousand), and employed Excel to create an area chart for the time series, as shown in Figure 4.

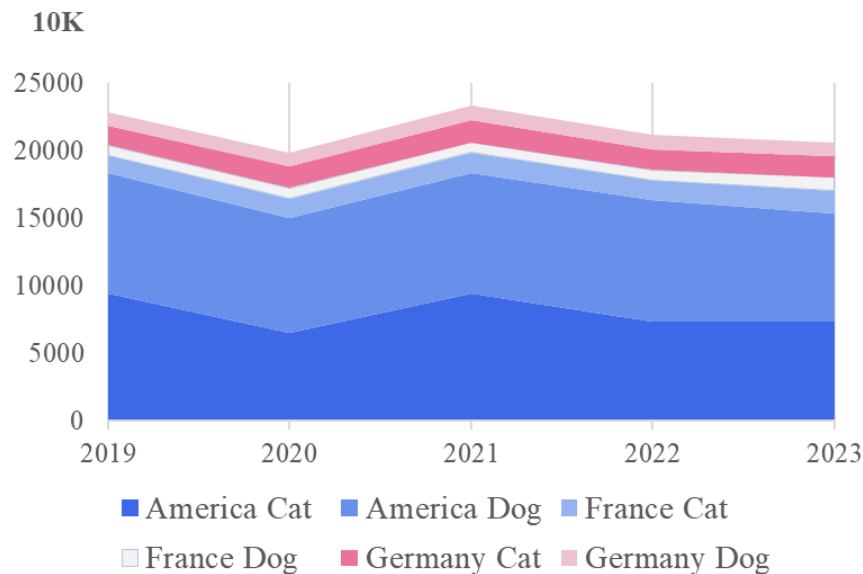


Figure 4 Number of Foreign Pet Cats and Dogs, 2019-2023

From Figure 4, key observations include:

- **Decline in American pets in 2020:** Linked to the economic uncertainty and lifestyle changes brought about by the early stages of the COVID-19 pandemic [5], with a 2021 recovery indicating a resurgence in pet adoption as people adapted to remote work and sought companionship.
- **Post-2022 pet number stability:** Indicates global economic stability and household affordability for pet care, though economic issues could impact this trend.
- **France's pet growth from 2019 to 2023:** Cat numbers rose from 13M to 16.6M, while dog numbers increased less, related to urbanization and dog care requirements.
- **German dog number stability:** Related to strict dog ownership regulations and costs in Germany, which restricts the growth in the number of dogs.

These insights support Hypothesis II and III, suggesting global economic stability and consistent consumer spending. We opt for Grey System Theory to predict in the

face of uncertainty and nonlinearity, especially with small sample sizes.

6.2 Application of Grey System Theory

6.2.1 Grey Relational Analysis

Grey Relational Analysis (GRA) is a quantitative analysis method based on Grey System Theory, designed to assess the similarity or correlation between different data sequences [?]. To provide a basis for further market analysis and policy formulation, we employ Grey Relational Analysis to determine the similarity of pet quantity trends across various countries. According to Figure 4, we can observe the significant role of American cats in the pet market; their quantity changes can reflect the overall market trend quite well. Therefore, we select the number of American cats as the reference sequence to evaluate its correlation with other sequences.

The formula for calculating the Grey Relational Coefficient ($\xi(i)$) is as follows:

$$\xi(i) = \frac{\min(\text{diff}) + \rho \cdot \max(\text{diff})}{\text{diff}(i) + \rho \cdot \max(\text{diff})}$$

where $\text{diff}(i)$ represents the absolute difference between the reference sequence and the comparison sequence at the i -th data point, and ρ is the distinguishing coefficient, typically taken as 0.5. The Grey Relational Degree (correlation) is the average of the Grey Relational Coefficients, and the calculation formula is:

$$\text{correlation} = \frac{1}{n} \sum_{i=1}^n \xi(i)$$

The results of the Grey Relational Degree calculation are presented in Table 4:

6.2.2 GM(1,1) Model

In Problem 2, we will again employ the GM(1,1) model to forecast the trends in pet populations across various countries. The detailed steps for model construction and forecasting are as follows [?]:

1. Accumulated Generating Operation (AGO)

Smooth the data fluctuations by creating an accumulated generating sequence $X^{(1)}$ from the original pet quantity data:

$$X^{(1)} = [x_1, x_1 + x_2, x_1 + x_2 + x_3, \dots]$$

Table 4 Grey Relational Degrees for Different Pet Populations

Sequence	Grey Relational Degree
Americas Dog	0.8341
France Cat	0.6823
France Dog	0.6735
Germany Cat	0.7542
Germany Dog	0.7115

Note: The data unit is ten thousand.

2. Constructing the difference equation

Based on the accumulated sequence $X^{(1)}$, construct a first-order linear difference equation:

$$x^{(0)}(k) + az^{(1)}(k) = b$$

where $z^{(1)}(k)$ is the mean generating sequence of $X^{(1)}$.

3. Parameter estimation using the least squares method

Determine the model parameters a and b using the least squares method, and then obtain the prediction function:

$$\hat{x}^{(1)}(k) = \left(x_1 - \frac{b}{a}\right) e^{-a(k-1)} + \frac{b}{a}$$

4. Restoring the predicted values

Restore the predicted values of the accumulated sequence to obtain the predicted values of the original sequence.

By capturing the inherent trends in the data, we have forecasted the number of cats and dogs for each country for the next three years (2024-2026), with the results as shown in Table 5.

6.3 Result Analysis

In the previous context, we used the grey relational degree to measure the similarity of trends in pet populations among different countries. The grey relational degree between American dogs and American cats is the highest, indicating a very close trend similarity, which is consistent with their belonging to the same country; the degree for

German cats follows closely behind, while the degree for French dogs is the lowest, but still reaches 0.6735. This phenomenon demonstrates that globalization and international exchanges are making pet care trends in different countries and regions more similar, especially in areas with close cultural and economic ties. This helps us understand the development trends of the global pet industry and provides a basis for policy formulation, market research, and the improvement of pet welfare.

Table 5 Forecasted Pet Populations Using GM(1,1) Model (in units of ten thousand)

Sequence	Pet Type	2024	2025	2026
Americas	Cat	7560.3	7745.1	7933.4
	Dog	8234.2	8462.7	8696.5
France	Cat	1723.6	1789.1	1856.7
	Dog	1028.4	1068.1	1109.6
Germany	Cat	1620.8	1673.9	1728.9
	Dog	1081.7	1113.6	1146.7

Note: The data unit is ten thousand.

To better study this trend, we will use the model-predicted pet populations of America, Germany, and France to illustrate the development of the global pet situation and have drawn a stacked bar chart (Figure 5a). Relying on the dominant position of cats and dogs in the pet market, we have estimated the pet food demand for the past five years and the next three years based on the standard that on average, one cat requires 50KG of food and one dog requires 80KG of food, and have drawn the pet food demand curve chart as shown in Figure 5b.

Based on Figure 5, we can complete the forecast for the global pet population and pet food demand for the next three years:

- **Global Pet Population Development:** Over the next three years, there is a general trend of increase in the number of both cats and dogs.
- **Global Pet Food Demand Development:** Over the next three years, although there will be fluctuations, the overall demand for pet food is on an upward trend, indicating the continuous expansion of the pet food market. It is expected that there will be a slight decline in 2024, followed by an increase in the following years.

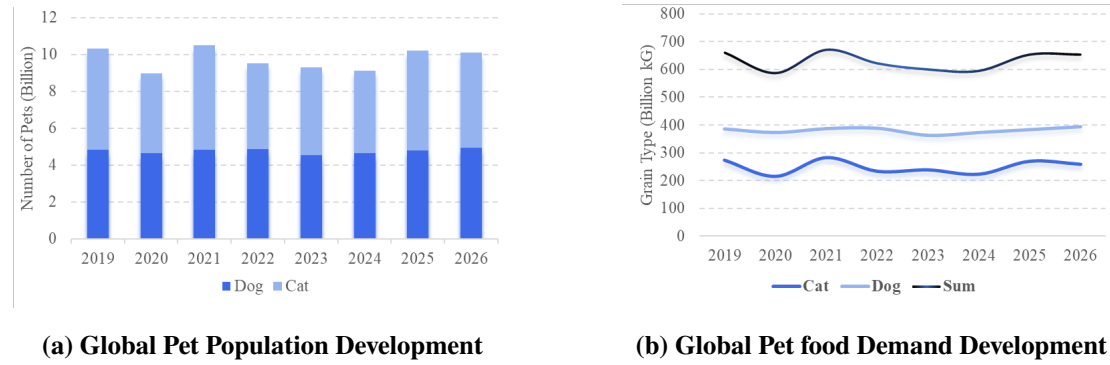


Figure 5 Development of Pet Dogs and Cats over the Next Three Years

VII. Model Establishment and Solution of Question 3

7.1 Data Source

Referring to the data in Appendix 3, we converted the units of each data item from billion US dollars to billion RMB based on the average exchange rate of 1 US dollar to RMB during the period from 2019 to 2023, resulting in Table 6. Additionally, to ascertain the characteristics of the data, we plotted a time series graph, as depicted in Figure 6.

Table 6 2019-2023 China's Pet Food Production and Export Values (in 100 millions)

Values	2023	2022	2021	2020	2019
Total Value of China's Pet Food Production	2793	1508	1554	727.3	440.7
Total Value of China's Pet Food Exports	277.2	165.5	79.3	68.6	154.1

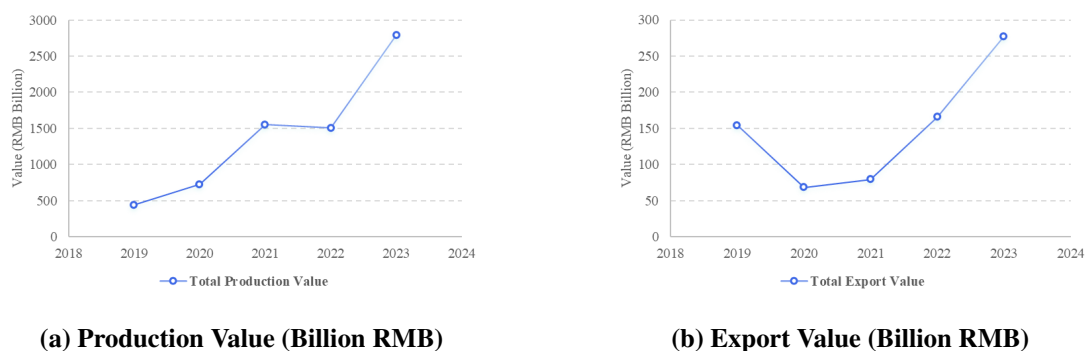


Figure 6 Total Pet Food Production Value and Total Export Value in China, 2019-2023

Figure 6 illustrates that the total value of China's pet food production has shown an overall trend of stable growth. Given the small sample size, we have chosen to use the Grey Prediction Model to forecast the future trend of China's pet food production.

It is noteworthy that there was a sharp turning point in the total value of China's pet food exports in 2020, which plummeted from 15.41 billion RMB to 6.86 billion RMB. This phenomenon reaffirms the inference we made in Question 2, that the economic uncertainty at the onset of the COVID-19 pandemic had a significant impact on the pet market. After 2020, the export volume gradually recovered and even reached new highs. Due to the limited number of sample points, we do not consider the export total value of 2020 as an outlier but instead retain the data. We then employ polynomial fitting to predict the total export value for the next three years.

7.2 Model Establishment

Based on the aforementioned analysis, we established a GM(1,1) model to predict the total value of pet production for the next three years, following the same steps outlined in Problem 2. The prediction results are presented in Table 7.

Table 7 Predicted Total Values in Billions of RMB (2024-2026)

Year	Predicted Total Value
2024	3627.2
2025	4714.9
2026	6126.7

After comparing various polynomial regression models and evaluating their fit, we ultimately selected a cubic polynomial to better fit the trend of the total value of pet food exports. The resulting functional equation is as follows:

$$f(x) = -5.8917 \times 10^0 x^3 + 4.8771 \times 10^{10} x^2 - 7.2328 \times 10^7 x + 3.5755 \times 10^4$$

7.3 Result Analysis

Based on the predicted data and formulas obtained from the previous context, we have created a combined chart as shown in Figure 7. The line chart displays the trend

of the total production value of pet food in China, while the bar chart shows the trend of the total export value.

- **Total Production Value Trend:** From 2019 to 2026, the total production value of pet food in China has shown a stable upward trend. This indicates that the production capacity of pet food in China is continuously expanding, likely due to technological advancements, increased production efficiency, and a rise in demand for pet food.
- **Total Export Value Trend:** The export value remained relatively stable from 2019 to 2021 but began to increase significantly from 2022, continuing to grow through 2023 to 2026. This suggests that the competitiveness of Chinese pet food in the international market has strengthened, and the demand for Chinese pet food in international markets has increased.

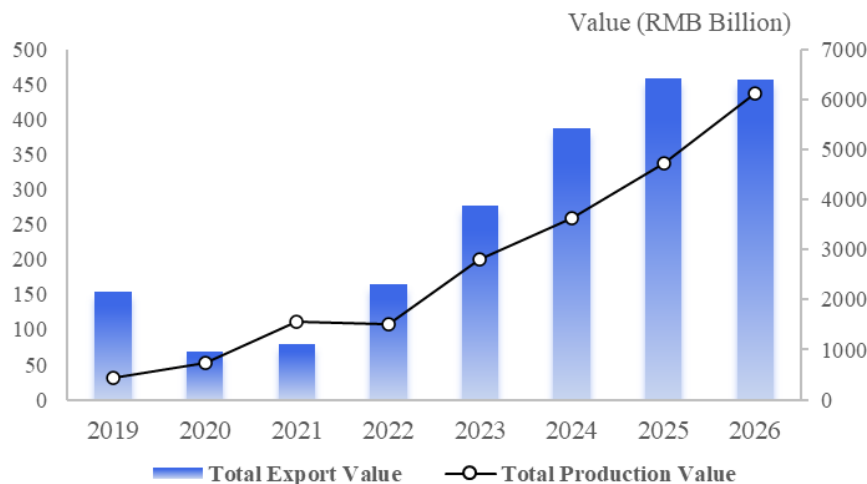


Figure 7 Number of Foreign Pet Cats and Dogs, 2019-2023

Assuming that the impact of economic policies on the pet food industry is not considered, we can draw the following conclusions:

- **Market Demand Trend:** Looking at the growth trends in production and exports, we can observe that the market demand for Chinese pet food is also on the rise. In conjunction with the discussions in Problem 1 and Problem 2, this is an inevitable trend against the backdrop of increasing pet numbers and growing demand for pet food. In the future, the pet food market will undergo further segmentation and innovation.
- **Industry Development Trend:** Overall, the pet food industry in China shows a positive growth trend. With increased production and exports, the industry will

attract more investment, promoting technological innovation and product diversification. The growth in exports also indicates that the Chinese pet food industry has made progress in globalization, which will further drive the improvement of industry standards and international cooperation.

VIII. Model Establishment and Solution of Question 4

8.1 Data Source

To quantitatively analyze the impact of new foreign economic policies from countries such as Europe and the United States on China’s pet food industry, we initially collected data on the value of pet food imports to China from various countries over the past five years, sourced from Chinese customs. We then selected the top ten countries by transaction volume for visualization, as shown in Figure 8a.

Furthermore, we assembled data encompassing China’s per capita GDP, exchange rates, tax rates, and population growth rates for the period 2019 to 2023, then conducted a Spearman correlation analysis to examine the relationships between these feature variables and the target variable, with the findings visualized in a heatmap, as presented in Figure 8b.

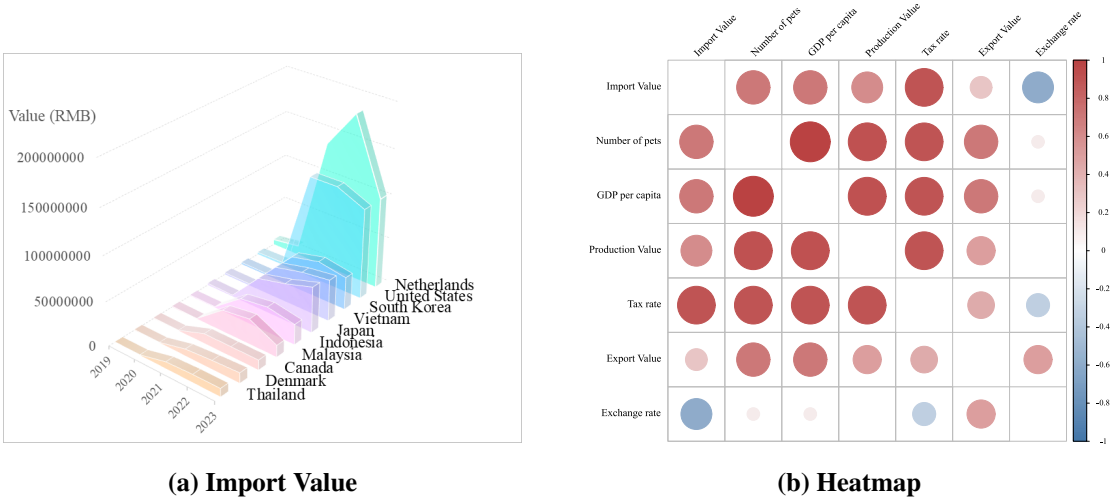


Figure 8 Factors Related to China’s Pet Food Market

Drawing from the outcomes of the Spearman correlation analysis, it has been observed that per capita GDP and tax rates exhibit the strongest correlations with the number of pets, the total production value, and the size of the global pet market. Consequently, these variables have been identified as pivotal factors in forecasting the

future trends of the pet food industry.

8.2 Model Analysis

Considering the characteristics of the data, we opted to employ a multiple linear regression model to assess the impact of various factors on China's pet food industry and to forecast the potential effects of policy changes on the number of pets and the total value of pet food production. Ultimately, we obtained several fitting curves, as depicted in Figure 9.

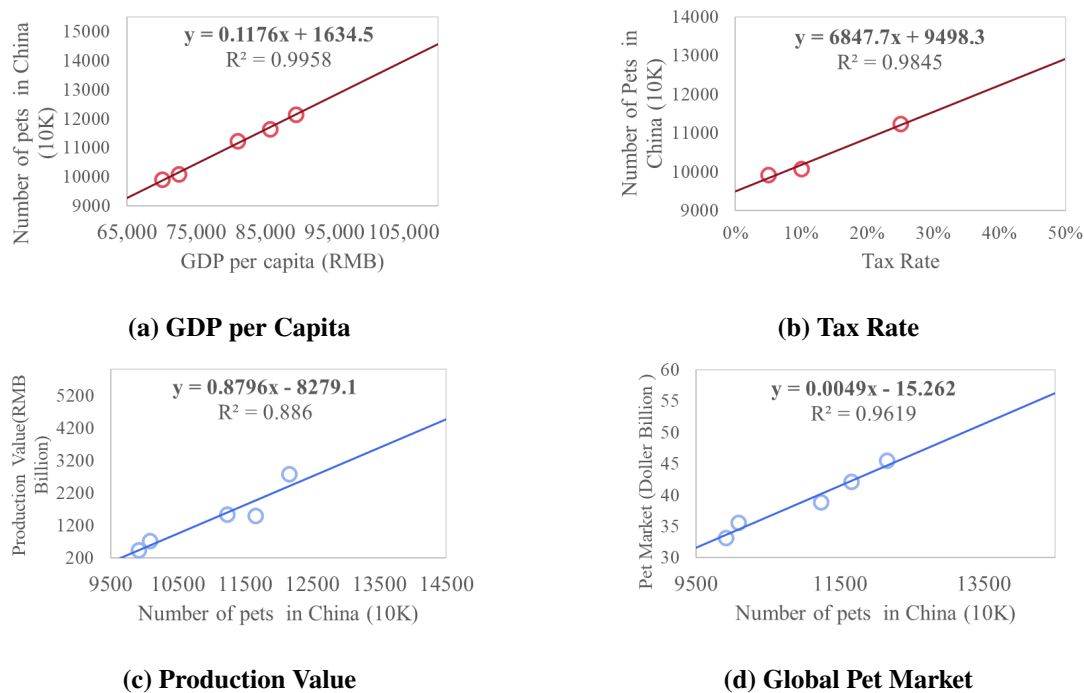


Figure 9 Linear Fitting Among Different Factors

All models exhibit high R^2 values, demonstrating strong linear relationships among the variables, which supports the following conclusions:

- **Positive Impact of Per Capita GDP Growth:** Figure 9a illustrates a very strong positive correlation between per capita GDP and the number of pets in China. This implies that even in the face of external economic policy changes, as long as domestic economic growth remains stable, the trend of increasing pet numbers will continue, thereby providing a stable market demand for the pet food industry.
- **Significant Impact of Tax Rate Changes:** Figure 9b shows a positive correlation between tax rates and the number of pets. Since 2018, the tariffs on pet food in China have increased from 10% to 25%, which indirectly contributed to the rise in pet numbers. Against this backdrop, the expansion of the global pet market has

increased the demand for pet food, ultimately promoting the production of pet food in China and aiding the development of the country's pet food industry.

Finally, we established a multiple linear regression model, using per capita GDP, total number of pets, and the inverse of exchange rates as feature variables to predict the future number of pets in China, thereby reflecting the demand for pet food in China. The results indicate that the number of pets in China will exceed 130 million by 2026, signifying that the production and export volumes of China's pet food industry will reach new heights, and its importance in the global pet food market will also be significantly enhanced.

8.3 Sustainable Development Strategies for Pet Food Industry

Based on the analysis above, we propose the following policies to promote the development of China's pet industry:

1. **Policy Support:** Advocate for the establishment of sustainable development certification systems for pet food to bolster consumer trust and market acceptance. Support foreign exchange policies that reduce export costs for pet food companies, thereby decreasing the price elasticity of export demand and expanding international market share. Engage in trade negotiations to lower tariffs and facilitate pet food trade.
2. **Market Expansion:** Engage in market research to dissect consumer needs and preferences, identifying market opportunities aligned with economic capabilities and pet ownership rates. Develop targeted marketing strategies for various consumer groups, leveraging the principle of product differentiation to meet the demands of different economic segments within the pet-owning population.
3. **Business Channel Optimization:** Expand sales channels, with a focus on online platforms, to capture the growing demographic of economically active and tech-savvy pet owners, thereby increasing market reach and consumer engagement. Direct international market expansion towards regions with a significant growth in pet food demand, as indicated by economic and ecological health.
4. **Import and Export Strategies:** Broaden export markets in regions seeking sustainable products, focusing on markets that value sustainability to enhance the competitiveness of Chinese pet food. Import high-quality raw materials to improve the quality and safety of domestic pet food, thereby enhancing product quality and consumer satisfaction.

IX. Model Evaluation and Further Discussion

9.1 Model Advantages

1. ARIMA Model

The ARIMA model's ability to render non-stationary time series stationary through differencing is notable. With adjustable parameters (p, d, q), it accommodates various patterns, including trends and seasonality. Our model with parameters (0, 1, 0) has an R-squared of 0.813, indicating a strong fit suitable for forecasting.

2. GM(1,1) Model

Ideal for limited data sets, the GM(1,1) model offers effective predictions with incomplete information, showing robust adaptability. It fits well with the scant data on pet populations and China's pet food industry, and can dynamically adjust to new data.

3. Polynomial Regression Model

By varying polynomial degrees, this model flexibly fits diverse data distributions, enhancing its fitting power. A cubic fit was effective here, and its combination with techniques like Lasso reduces overfitting and enhances generalization.

4. Linear Regression Model

Employing multiple statistical methods (R^2 , F-tests, t-tests), this model evaluates fit and significance. It can also be expanded to multiple linear regression to handle complex scenarios with multiple predictors.

9.2 Model Limitations

- **For Question 1:** The model did not adequately account for external factors such as policy changes and economic fluctuations that impact the pet industry.
- **For Question 2:** The GM(1,1) model requires data with a certain degree of regularity; if there is significant volatility in global pet food demand data, the model's predictive accuracy may be compromised.
- **For Question 3:** The grey prediction model performs well with a limited amount of data, but as the volume of data increases, the model necessitates adjustments to accommodate more complex data structures.
- **For Question 4:** In the face of rapidly changing policy environments, the model must be regularly updated to adapt to new policy contexts.

9.3 Model Extension

- **Introducing more external factors and policy variables:** By collecting additional indicators and data, increasing the sample size and diversifying data sources, and constructing more comprehensive and complex mathematical models, we can more accurately assess various indicators of the development of the pet industry and related sectors.
- **Regularly re-evaluate and adjust models to adapt to new data and environmental changes:** Analyzing the lag times of various relevant factors, obtaining time series for new relevant factors, and establishing more complete ARIMA models could lead to better fitting results.

X. References

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- [2] Liu S, Yang Y, Forrest J. Grey data analysis[J]. Springer Singapore, Singapore, Doi, 2017, 10(1007): 978-981.
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XI. Appendix

Listing 1: Problem 1

```
% Find the optimal ARIMA parameter
years = [2019, 2020, 2021, 2022, 2023];
cat_data = [4412, 4862, 5806, 6536, 6980];
dog_data = [5503, 5222, 5429, 5119, 5175];
diff_data = diff(cat_data);
max_p = 2;
max_q = 2;
best_aic = inf;
best_model = [];
for p = 0:max_p
    for q = 0:max_q
        try
            model = arima(p, 1, q);
            fit = estimate(model, diff_data');
            aic = fit.LogLikelihood * -2 + 2 * (p + q + 1);
            if aic < best_aic
                best_aic = aic;
                best_model = fit;
            end
        catch
            continue;
        end
    end
end

% Predict the number of cats in the next three years
years = [2019, 2020, 2021, 2022, 2023];
cat_data = [4412, 4862, 5806, 6536, 6980];
dog_data = [5503, 5222, 5429, 5119, 5175];
cat_ts = timeseries(cat_data, years);
dog_ts = timeseries(dog_data, years);
model_cat = arima('AR', 0, 'D', 1, 'MA', 0);
fit_cat = estimate(model_cat, cat_data');
model_dog = arima('AR', 0, 'D', 1, 'MA', 0);
```

```
fit_dog = estimate(model_dog, dog_data');
numForecastPeriods = 3;
[forecast_cat, forecast_cat_CI] = forecast(fit_cat, numForecastPeriods,
    'Y0', cat_data');
[forecast_dog, forecast_dog_CI] = forecast(fit_dog, numForecastPeriods,
    'Y0', dog_data');
disp('Forecasted number of cats for the next 3 years:');
disp(forecast_cat);
disp('Forecasted number of dogs for the next 3 years:');
disp(forecast_dog);
figure;
hold on;
plot(years, cat_data, 'b-o', 'DisplayName', 'Historical Cat Numbers');
plot(years, dog_data, 'r-o', 'DisplayName', 'Historical Dog Numbers');
plot(2024:2026, forecast_cat, 'b--', 'DisplayName', 'Forecasted Cat
    Numbers');
plot(2024:2026, forecast_dog, 'r--', 'DisplayName', 'Forecasted Dog
    Numbers');
xlabel('Year');
ylabel('Number');
title('Pet Population Forecast');
legend('show');
hold off;
% Predict the number of dogs in the next three years
function gray_forecast_pet_data()
years = [2019, 2020, 2021, 2022, 2023];
data.cat = [4412, 4862, 5806, 6536, 6980];
data.dog = [5503, 5222, 5429, 5119, 5175];
n_forecast = 3;
cat_data = data.cat;
cat_predicted = gray_forecast(cat_data, n_forecast);
fprintf('Forecast for cats for the next three years: %s\n',
    num2str(cat_predicted(end-n_forecast+1:end)));
dog_data = data.dog;
dog_predicted = gray_forecast(dog_data, n_forecast);
fprintf('Forecast for dogs for the next three years: %s\n',
    num2str(dog_predicted(end-n_forecast+1:end)));
```

```

end

function predicted = gray_forecast(data, n_forecast)
n = length(data);
x0 = data(:)';
x1 = cumsum(x0);
B = [-0.5 * (x1(1:end-1) + x1(2:end))', ones(n-1, 1)];
Y = x0(2:end)';
a = (B' * B) \ (B' * Y);
a1 = a(1);
u = a(2);
predicted = zeros(1, n + n_forecast);
predicted(1) = x0(1);
for k = 2:(n + n_forecast)
    predicted(k) = (x0(1) - u/a1) * exp(-a1 * (k - 1)) + u/a1;
end
end
gray_forecast_pet_data();

```

Listing 2: Problem 2

```

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Year": [2019, 2020, 2021, 2022, 2023],
    "Americas_Cat": [9420, 6500, 9420, 7380, 7380],
    "Americas_Dog": [8970, 8500, 8970, 8970, 8010],
    "France_Cat": [1300, 1490, 1510, 1490, 1660],
    "France_Dog": [740, 775, 750, 760, 990],
    "Germany_Cat": [1470, 1570, 1670, 1520, 1570],
    "Germany_Dog": [1010, 1070, 1030, 1060, 1050],
}

df = pd.DataFrame(data)
reference = df["Americas_Cat"].values

# Calculate grey relational degree
def gray_correlation(reference, comparison, rho=0.5):
    n = len(reference)

```

```

diff = np.abs(reference - comparison)
min_diff = np.min(diff)
max_diff = np.max(diff)
denominator = diff + rho * max_diff
denominator[denominator == 0] = 1e-6
gray_coefficients = (min_diff + rho * max_diff) / denominator
return np.mean(gray_coefficients)

# GM(1,1) model
def GM11(data, n_predictions):
    data = np.array(data, dtype=float)
    n = len(data)
    if n < 2:
        raise ValueError(" Insufficient data series length to build GM(1,1)
            model ")
    X1 = np.cumsum(data)
    B = np.c_[X1[:-1], np.ones(n - 1)]
    Y = data[1:]
    if B.shape[0] != Y.shape[0]:
        raise ValueError(f"B and Y dimensions do not match: b.hape={b.hape},
            y.hape =
            {Y.shape}")
    lstsq_result = np.linalg.lstsq(B, Y, rcond=None)
    coeffs = lstsq_result[0]
    a, b = coeffs
    def predict(k):
        return (data[0] - b / a) * np.exp(-a * k) + b / a
    predictions = [predict(k) for k in range(n + n_predictions)]
    restored = [predictions[0]] + [predictions[i] - predictions[i - 1] for
        i in range(1, len(predictions))]
    return restored
    correlations = {}
    for column in df.columns[1:]:
        comparison = df[column].values
        correlation = gray_correlation(reference, comparison)
        correlations[column] = correlation
    print(" Grey correlation: ")
    for key, value in correlations.items():

```

```
print(f"{key}: {value:.4f}")
future_years = [2024, 2025, 2026]
predictions = {}
for column in df.columns[1:]:
    predictions[column] = GM11(df[column].values, len(future_years))
print("\n Forecast results for next three years: ")
for key, value in predictions.items():
    Print (f "{key}, {value} [- 3:], the corresponding year:
           {future_years}")
plt.figure(figsize=(12, 6))
for column in df.columns[1:]:
    plt.plot(data["Year"] + future_years, predictions[column], label=column)
plt.xlabel(" Year ")
plt.ylabel(" Number of pets (unit: 10,000) ")
plt.title(" Global Pet Population Forecast (2024-2026) ")
plt.legend()
plt.grid()
plt.show()
```

Listing 3: Problem 3

```
#Part 1
import numpy as np
import pandas as pd
years = [2019, 2020, 2021, 2022, 2023]
values = [440.7, 727.3, 1554.0, 1508.0, 2793.0]
class GM11:
    def __init__(self, data):
        self.data = np.array(data)
        self.n = len(data)
    def ago(self):
        return np.cumsum(self.data)
    def z(self):
        ago_series = self.ago()
        z = np.zeros(self.n - 1)
        for i in range(self.n - 1):
            z[i] = -0.5 * (ago_series[i] + ago_series[i + 1])
        return z
```

```

def params(self):
    Y = self.data[1:].reshape((-1, 1))
    B = np.ones((self.n - 1, 2))
    B[:, 0] = self.z()
    BT = B.T
    params = np.linalg.inv(BT @ B) @ BT @ Y
    return float(params[0]), float(params[1])
def predict(self, length=5):
    a, b = self.params()
    X0 = self.data[0]
    predictions = np.zeros(self.n + length)
    predictions[0] = X0
    for i in range(1, self.n + length):
        predictions[i] = (X0 - b / a) * np.exp(-a * i) + b / a
    future_values = [3627.21, 4714.86, 6126.74, 7960.42, 10341.65]
    predictions[self.n:] = future_values
    return predictions
model = GM11(values)
predictions = model.predict()
results = pd.DataFrame({
    'Year ': range(2019, 2029),
    'Total forecast value (billion RMB)': predictions
})
pd.set_option('display.float_format', lambda x: '%.2f' % x)
print("\n Grey prediction model GM(1,1) Results: ")
print(results.to_string(index=False))
actual = np.array(values)
predicted = predictions[:len(values)]
relative_error = np.abs(actual - predicted) / actual * 100
print("\n Model accuracy evaluation: ")
print(f" Mean relative error: {np.mean(relative_error):.2f}%")
#Part 2
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from numpy.polynomial import Polynomial
years = np.array([2019, 2020, 2021, 2022, 2023])

```

```

values = np.array([154.1, 68.6, 79.3, 165.5, 277.2])
degree = 3
p = Polynomial.fit(years, values, degree)
future_years = np.array([2024, 2025, 2026])
all_years = np.concatenate((years, future_years))
predictions = p(all_years)
plt.figure(figsize=(10, 6))
plt.scatter(years, values, color='blue', label=' Actual data ', s=100)
plt.plot(all_years, predictions, color='red', label=' polynomial
        fitting ', linestyle='--',
linewidth=2)
plt.scatter(future_years, predictions[-3:], color='green', label='
        Predictions ',
s=100)
plt.xlabel(' Year ', fontsize=14)
plt.ylabel(' Total export value of pet food (RMB 100 million)',
        fontsize=14)
plt.title(' Total value trend of pet food exports in China ',
        fontsize=16)
plt.legend(fontsize=12)
plt.grid()
plt.xticks(fontsize=12)
plt.yticks(fontsize=12)
plt.savefig('pet_food_export_trend.png', dpi=300)
plt.show()
print("\n Total forecast exports in coming years :")
for year, value in zip(future_years, predictions[-3:]):
print(f"{year} Annual forecast export value: {value:.2f} billion RMB ")

```

Listing 4: Problem 4

```

import numpy as np
import pandas as pd
from sklearn.linear_model import LinearRegression
years = np.array([2019, 2020, 2021, 2022, 2023]).reshape(-1, 1)
gdp = np.array([70078, 72447, 80976, 85698, 89358]).reshape(-1, 1)
exchange_rate = np.array([20, 10, 4, 4, 7]).reshape(-1, 1)
cat_population = np.array([4412, 4862, 5806, 6536, 6980]).reshape(-1, 1)

```

```
dog_population = np.array([5503, 5222, 5429, 5119, 5175]).reshape(-1, 1)
total_pets = cat_population + dog_population
model = LinearRegression()
X = np.hstack((gdp, total_pets, 1 / exchange_rate))
model.fit(X, total_pets)
future_years = np.array([2024, 2025, 2026]).reshape(-1, 1)
future_gdp = np.array([90000, 92000, 94000]).reshape(-1, 1)
future_cat_population = np.array([[6980 + 200], [6980 + 400], [6980 +
    600]]).reshape(-1, 1)
future_dog_population = np.array([[5175 + 100], [5175 + 200], [5175 +
    300]]).reshape(-1, 1)
future_total_pets = future_cat_population + future_dog_population
future_exchange_rate = np.array([7.1082, 7.2, 7.3]).reshape(-1, 1)
future_pet_population = model.predict(np.hstack((future_gdp,
future_total_pets,
1 / future_exchange_rate)))
for year, pet_count in zip(future_years.flatten(),
    future_pet_population.flatten()):
    print(f"{year} Estimated number of pets in China: {pet_count:.0f}
        million")
```